



The influence of persona and conversational task on social interactions with a LLM-controlled embodied conversational agent

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ABSTRACT

Large Language Models (LLMs) have demonstrated remarkable capabilities in conversational tasks. Embodying an LLM as a virtual human allows users to engage in face-to-face social interactions in Virtual Reality. However, the influence of person- and task-related factors in social interactions with LLM-controlled agents remains unclear. In this study, forty-six participants interacted with a virtual agent whose persona was manipulated as “extravert” or “introvert” in three different conversational tasks (small talk, knowledge test, convincing). Social-evaluation, emotional experience, and realism were assessed using ratings. Interactive engagement was measured by quantifying participants’ words and conversational turns. Finally, we measured participants’ willingness to ask the agent for help during the knowledge test. Our findings show that the extraverted agent was more positively evaluated, elicited a more pleasant experience and greater engagement, and was assessed as more realistic compared to the introverted agent. Whereas persona did not affect the tendency to ask for help, participants were generally more confident in the answer when they had help of the LLM. Variation of personality traits of LLM-controlled embodied virtual agents, therefore, affects social-emotional processing and behavior in virtual interactions. Embodied virtual agents allow the presentation of naturalistic social encounters in a virtual environment.

1. Introduction

The rise of large language models (LLMs) has revolutionized human-computer interaction, enabling more sophisticated and intuitive exchanges between people and machines. LLMs can be instated as conversational agents (e.g. ChatGPT), allowing users to engage in meaningful language-based interactions. While earlier versions of conversational agents were based on simple pattern-matching algorithms that generated pre-defined output (Ramesh et al., 2017), LLMs use probabilistic generative processes based on training data to generate responses (Generative Pre-Trained Transformer, GPT; Yenduri et al., 2024). Furthermore, because LLMs can consider the contextual information of an input, they allow the generation of individualized responses that match the user’s intention. Because of these properties LLM controlled conversational agents are increasingly being used in chat-based interactions for example in customer service and healthcare (Rivas & Zhao, 2023; Thirunavukarasu et al., 2023). However, while

chat-based interactions are useful in many tasks, face-to-face conversations may be advantageous as they allow the simulation of ecological valid human-to-human interactions and the presentation of additional information via non-verbal cues (e.g., gesture, gaze, facial expressions). Importantly, face-to-face social interactions are better suited for establishing a deeper social connection between interactive partners than text based interactions (Sacco & Ismail, 2014). This has inspired the development of embodied conversational agents, in which computer-based dialogue systems are combined with animated virtual agents (Huang, 2018). Presenting embodied conversational agents in Virtual Reality (VR) as 3D, life-like virtual humans enable users to engage in multi-modal, face-to-face conversation with an interactive partner in front of them. As with chatbots, recent attempts have been made to increase the naturalism of conversations by incorporating LLMs as embodied conversational agents (Hasan et al., 2023; Lim et al., 2024). The high degree of flexibility and individualization of LLM controlled conversations makes them superior to scripted speech stimuli implemented in

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virtual scenarios that can only follow a pre-defined conversational path. Another benefit of LLMs is that they are pretrained for standard conversation in natural language. Without the need to collect huge, but specialized data corpora for training dialogue models, LLMs can be integrated immediately in virtual agents. This allows use cases in research focused on studying basic mechanisms of social interactions (Pan & Hamilton, 2018) and supports practical applications, such as conversations with virtual patients in medical education (Graf et al., 2024), or natural social scenarios in virtual exposure therapy. However, while such set-ups can have a profound impact on human experience and behavior, little is known about the underlying mechanisms and contextual influences that drive social interactions with LLM controlled embodied virtual agents.

To implement social interactions with LLM-controlled embodied virtual agents, it is crucial to determine if interactions adhere to the same principles as human-human interaction, affecting socio-emotional evaluation and behavior. According to the computers as social actors (CASA) framework persons automatically respond to social cues in an interaction regardless of whether these cues are produced by a human or computer (Isbister & Nass, 2000; Nass & Moon, 2000). This effect can be enhanced through anthropomorphic visual cues (Kim & Sundar, 2012) or behavioral realism, i.e. the degree to which an agent's social cues and behavior resemble human ones (Von Der Pütten et al., 2010). The CASA framework shows that people form social evaluations in interactions with virtual agents (Guadagno et al., 2011) and chatbots (Liu & Sundar, 2018), similar to human interactions (Bar et al., 2006; Satchell, 2019). Attributing social traits like personality to an interactive agent can influence a person's social and emotional processing of the interactive partner. For instance, in text-based communication an agent's assertive and serious personality compared to a warm and cheerful personality increased the willingness to confide in the agent during a job interview (Zhou et al., 2019). With regard to specific personality dimensions of conversational agents it was shown that chatbots scoring high on extraversion resulted in higher social presence and increased communicative satisfaction compared to less extraverted chatbots (Ahmad et al., 2021). LLMs can effectively incorporate specific personality profiles (Frisch & Giulianelli, 2024; Jiang et al., 2024; Serapio-García et al., 2023). Combined with embodied conversational agents, such implementations may strongly affect users experience (Lim et al., 2024; Sonlu et al., 2024). Importantly, social interactions influence not only social evaluation but also behavior. Mimicry, i.e. the imitation of behavior by an observer (Chartrand & Lakin, 2013), is thought to increase affiliation between persons (Lakin & Chartrand, 2003; Salazar Kämpf et al., 2018). While studies have often focused on mimicry of non-verbal behavior, similar claims have been made for verbal behavior in the interactive alignment model (Pickering & Garrod, 2004). LLM-controlled embodied conversational agents allow to investigate verbal alignment in interactions with a virtual agent. Overall, the communicative style and personality traits of a conversational agent determine user experience in interactions. However, this has been mostly tested in text-based communication or with pre-determined responses, while data on personality effects in LLM-controlled agents presented face-to-face as embodied interactive partners in Virtual Reality remains sparse.

Conversational agents built on LLMs can be used for a wide variety of tasks. They can be used to answer specific questions about world knowledge, provide detailed instructions about how to perform a particular task, give everyday advice, or serve as social companions in small-talk or even deeper conversations (Skjuve et al., 2023). Importantly, the nature of a task or context may influence how users interact with a conversational agent. For instance, receiving help from a conversational agent may generally lead to a more pleasant experience. Furthermore, tasks may interact with personality traits or communicative style, in a sense that particular traits might be positively evaluated in some tasks but not in others. Only few studies have investigated this relationship. Roy et al. (2021) found that extraverted agents were more favorably rated than introverted agents in tasks where the goal was to

provide information, whereas there were no clear preferences in tasks where the goal was to complete an assignment. Another study demonstrated that users' competency levels may also play a role in this regard, as high-competency users rated a social-oriented interactive style as more useful, while low-competency users rated a task-oriented style as more useful when interacting with a conversational agent (Chattaraman et al., 2019). Taken together, tasks may play an important role in how user's form social evaluations about conversational agents; however, studies that investigate LLM-controlled embodied virtual agents across different tasks are missing.

While the CASA framework has been around for several years it has been mostly tested with respect to changes in attitudes towards computers (Nass & Moon, 2000) and has not been directly tested with respect to LLM controlled virtual agents. Importantly, because interactions with LLM-controlled agents allow for more complex and plausible social encounters, more sophisticated social effects (as predicted by interactive alignment model) as well as context effects can be investigated and context effects need to be evaluated. The goal of the present study is thus to investigate the experience and behavioral adaptation in plausible conversations with LLM-controlled virtual agents. More specifically, the study was designed to test how persona descriptions of an LLM-controlled virtual agent and conversational tasks would affect experience and behavior in face-to-face social interactions. Participants engaged in face-to-face social interactions with an LLM-controlled agent in Virtual Reality. The persona of the LLM was manipulated to be either "extraverted" or "introverted". Three conversational tasks were conducted: Small Talk, Knowledge Test, and Convincing. Tasks were chosen to reflect a social conversation focused on self-disclosure (small talk), task-oriented conversation (knowledge test), and social conversation focused on argumentation (convincing). Ratings were assessed to measure the social evaluation of the agent (sympathy and closeness), emotional experience (valence and arousal), and the experience of the virtual interaction (realism and social presence). In addition, the number of words and turns as well as the number of times participants asked the agent for help in the knowledge test were measured to quantify interactive engagement.

Based on previous findings a set of hypotheses was preregistered. (1) First, because small talk is typically experienced as pleasant even when conducted with strangers (Epley & Schroeder, 2014), and because it may establish intimacy during self- and other disclosure (Park et al., 2011) we expected that small talk conversation would be experienced more pleasant than the knowledge test and the argumentation during the convincing task. (2) The CASA framework predicts that participants would act towards the virtual agent as they would towards another human (Nass & Moon, 2000). Consequently, they should be sensitive to the manipulation of personality with an impact on social evaluation and emotional experience. As previous studies found a benefit for social and extraverted agents (Chattaraman et al., 2019; Roy et al., 2021), we expected that the evaluation of the agents persona would differ as a function of conversational task with respect to sympathy, valence, closeness, and realism, with higher ratings for the extraverted compared to the introverted agent. This difference was expected to be more pronounced in the small talk than in the knowledge and convincing task. (3) In this line, given that participants acted with the virtual agent as with a human, this should also influence their behavior. According to the interactive alignment framework persons align their behavioral patterns to their interactive partners during interactions (Chartrand & Lakin, 2013; Pickering & Garrod, 2004). One major aspect of the behavior of interest in the present study was the quantity of verbal behavior, as the extraverted agent was manipulated to be more talkative than the introverted agent. As predicted by the interactive alignment model for human-human interactions and adapted to human-computer interactions by the CASA framework, participants should then exhibit aligned behavioral patterns, i.e. being more talkative when interacting with the talkative, extraverted compared to the less-talkative, introverted agent. However, because such imitative behavior has been

linked to affiliative goals (Hess, 2021) this should be more prominent in the small talk task. Therefore we hypothesized an interaction effect between persona and conversational tasks on interactive engagement, i.e. the number of words spoken by the participant during a conversation: more words were expected in the extraverted compared to introverted condition. This difference was expected to be greater in the small talk task than in the knowledge and convincing tasks. (4) Finally, a more positive social evaluation through the personality manipulation might also be evident in behavior towards the agent, i.e. pleasant social interactions with a talkative and social extraverted agent might lower the barrier to ask for help in the knowledge task (Ashfaq et al., 2020). Consequently, it was hypothesized that participants would ask the extraverted agent more frequently for help than the introverted agent.

2. Methods

2.1. Participants

Forty-six healthy volunteers were recruited at Regensburg University (mean age = 21.24 years, SD = 2.59, 36 females). Participants were mostly recruited via advertisement at the university and the sample consisted of 96 % students from which 95 % studied in the psychology undergrad program. All participants were German-native speakers and had a Western European cultural background. Participants were randomly assigned into two groups of 23 participants with groups being matched for gender. A sensitivity analysis conducted using G*Power (Faul et al., 2009) revealed that this sample size allowed the detection of a medium to large effect size of $\eta_p^2 = .10$ with a power of .8 for a mixed effect ANOVA design. Participants did not report mental or neurological disorders and had normal or corrected-to-normal vision. All participants gave written informed consent. The study was conducted in line with the Declaration of Helsinki and was approved by the ethics committee of Regensburg University (24-3669-101). Participants received credit points for compensation.

2.2. Experimental design

The study used a mixed design with the agent's *persona* (extraverted vs. introverted) as a between-subjects factor and *conversational task* (small talk vs. knowledge test vs. convincing) as a within-subject factor. During the knowledge test an additional manipulation was introduced by alternating between *easy* and *difficult* questions that had to be

answered by the participants. Sympathy, Valence, Arousal, Closeness, and Naturalism were measured via self-report after each conversational task and number of words and turns during each task were assessed as behavioral measures. In the knowledge test, the percentage of questions for which the participants asked the agent for help was assessed as an additional metric. Finally, the social presence subscale of the multi-modal presence scale (MPS, Makransky et al., 2017) was measured for each conversational task.

2.3. Apparatus

2.3.1. Technical set-up

All system components are shown in the schematic overview in Fig. 1. Virtual Reality was presented via a head-mounted display (Vive Pro, HTC, Taoyuan, Taiwan). The virtual environment was rendered and controlled using Unreal Engine 5.2 (Epic Games Inc., Raleigh, USA). Participants were placed in a virtual room at a table face-to-face with a male virtual agent (MetaHuman, Epic Games Inc.). The same agent was used in both between-subject personality conditions and the distance between participant and agent was set to 1.8 m. Eye gaze of the virtual agent was controlled so that the agent was always looking towards the participant. Participant speech was sampled with an external USB microphone (t-bone SC 420, Thomann, Germany) and converted to text using the wav2vec model (Baevski et al., 2020). The text input was then forwarded as a prompt to a text generation model ("SauerkrautLM-HerO", <https://huggingface.co/VAGOSolutions/SauerkrautLM-7b-HerO>) which was run on a separate computer using the text generation inference toolkit with streaming enabled. The transcribed text was also written to a file for the analysis of participant's speech. The final prompt forwarded to the model concatenated the persona prompt of the respective experimental condition and the input from the participants. The prompt template included labels for the system prompt (<|im_start|>system *Persona Text* <|im_end|>), the user input (<|im_start|>user *Input from User* <|im_end|>), and a non-closed LLM response prompt in the final line (<|im_start|>NAME_AGENT) which resulted in a response generation. We did not include the history of previous exchanges. Importantly, deploying all components (including the LLM) on local servers allowed us to maintain full control over all data without the need to rely on third-party services and in accordance to data protection policies. Across all experimental sessions, the average time of the LLM to generate a response was 1.29 s (SD = .26, range = .68–4.11 s).

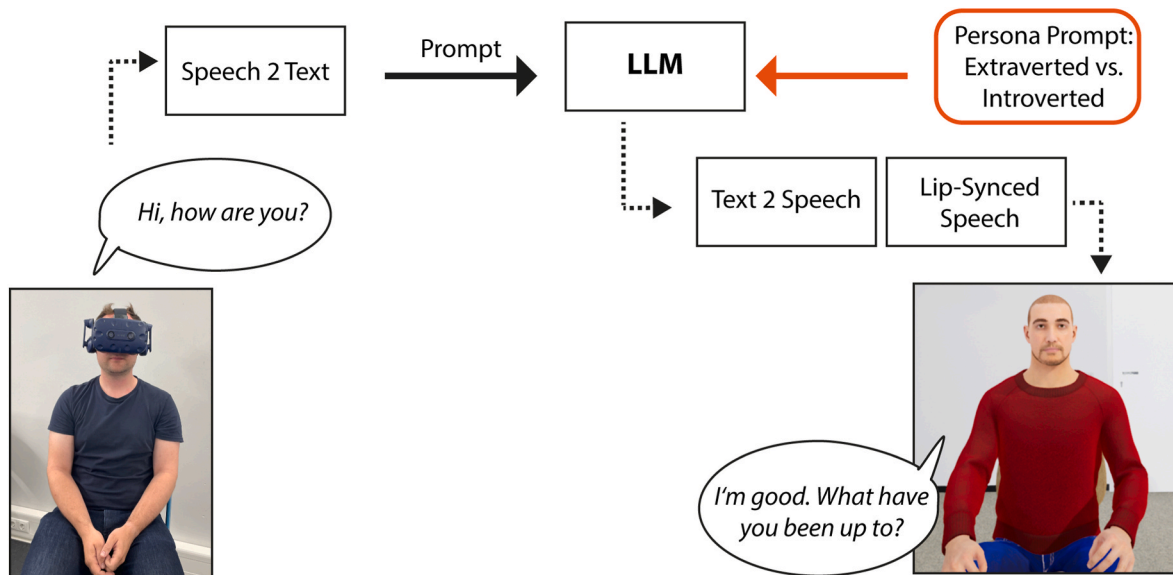


Fig. 1. Schematic overview of system components.

Two different personas were implemented as context prompts to the text generation model, i.e. prompts that preceded the user input, depending on the persona manipulation “extravert” or “introvert”. The personas included details about the communicative style and gave examples on how the agent would answer particular questions (see Table 1, full text in Supplementary Material). The descriptions varied in how persona were described as being sociable, talkative, and enjoying engaging with other people which all relates to the dimension of extraversion in the five-factor theory of personality (McCrae & John, 1992). Each persona also included the same information about how the LLM had to answer specific questions about preferences. This information was included for two reasons: First, to ensure consistency in conversations so that the LLM model would not change preferences or biographical information within a conversation. And second, to ensure that no differences in sympathy based on preferences (e.g. liking dogs better than cats) could arise between personas. Additional parameters, such as the maximum number of generated sentences and the temperature parameter, were manipulated between the personas (Table 1). The parameters were tested by the experimenters before the start of the experiment to ensure that the LLM would produce realistic and consistent responses within a conversation. The text output of the LLM was then converted into speech (male voice) using a speech2text model (Silero, <https://github.com/snakers4/silero-models>). The text output of the LLM was also written to a logfile. The same voice was used for all personality conditions. Finally, the resulting audio was streamed to the Audio2 Face application (Nvidia Omniverse, <https://build.nvidia.com/nvidia/audio2face>) where live lip movements were animated based on the “Mark” set-up. Animations and audio were then played by the virtual agent using the LiveLink plugin in the Unreal Engine.

2.3.2. Ratings and questionnaires

After each conversational task, participants were asked to rate their experience using a visual analog scale (values 0–100). Ratings included dimensions to describe emotional experience (valence and arousal, Russell, 1980), as well as social evaluation of the virtual agent (sympathy and closeness; Gruen & Mendelsohn, 1986; Rychlowska et al., 2017). These ratings included sympathy (“How sympathetic was the virtual person?”, 0 = very unsympathetic, 100 = very sympathetic), valence (“How unpleasant or pleasant did you feel during the interaction?”, 0 = very unpleasant, 100 = very pleasant), arousal (“How high was your arousal during the interaction”), closeness (“How close/connected did you feel to the virtual person?”, 0 = very distanced/unconnected, 100 = very close/connected), realism (“How unnatural or natural did you experience the interaction?” 0 = very unnatural, 100 = very natural). Because the “knowledge test” conversational task required participants to answer questions about general world knowledge, two additional ratings were included that asked for participants’ confidence in the correctness of their answer for questions answered with or without help of the virtual agent (“How confident are you that your answers were correct for

Table 1
Short persona description and LLM parameters. For full persona descriptions see supplementary material.

Persona	Prompt	Temperature	Maximum Sentences
Extravert	Your name is Moritz. You have a very extroverted character. This means you are very sociable, enjoy being around people, are active, and talkative. You engage a lot with your conversation partner and ask interested follow-up questions. [...] }	.4	3
Introvert	Your name is Moritz. You are a very introverted character. This means you enjoy being alone, are independent, and very reserved in social interactions. [...]	.2	2

questions that you answer on your own/for questions that you answered with help of the virtual person?”, 0 = very unconfident, 100 = very confident).

Social anxiety symptoms were assessed using the Social Phobia Inventory (SPIN, Connor et al., 2000). In addition, the social presence subscale of the Multimodal Presence Scale (MPS; Makransky et al., 2017) was used to measure the experience of social presence in each conversational task. The subscale averages over five items that have to be answered on a 5-point Likert scale. Furthermore, a German translation of the General Attitudes towards Artificial Intelligence Scale was used (GAAIS, Schepman & Rodway, 2020). The questionnaire included 16 positive items (e.g. opportunities, benefits, and positive emotions related to AI) and 16 negative items (concerns and negative emotions related to AI). Items were answered on a 5-point Likert scale, and average scores were calculated for each subscale. Finally, personality traits were measured using the 30 item short version of the NEO five-factor-inventory (NEO-FFI-30, Körner et al., 2008). Data from questionnaires are accessible in the open data but were not analyzed in the present study.

2.4. Procedure

Participants were asked to provide informed consent, fill in questionnaires about demographic information (age, sex, and occupation) and social anxiety symptoms. This initial part had a duration of approximately 15 min. Participants then sat on a chair and put-on the HMD where the virtual room was presented. Participants position in the virtual room was adjusted so that they were sitting at the virtual table, directly facing the virtual agent. The virtual agent was presented from the beginning, but the LLM was not activated before the conversational task, i.e. there was no speech by the agent before the first conversational task.

Next, the first conversational task was initiated by the experimenter. The small talk conversation was always implemented as the first task in order to simulate a natural interaction in which persons first get to know each other. Instructions about the task were displayed on a virtual screen on the table and participants used the motion controller to forward through instructions. Participants were informed that they would be prompted to verbally ask the virtual agent small talk questions displayed on the virtual screen and that they could use the motion controller to forward to the next question. Participants were also specifically instructed that they were free to follow-up on anything the virtual agent said (literal translations of all instructions are presented in the Supplementary Material). The conversational task was then conducted for a total of 8 min, during which the LLM was set active so that responses were generated for any input that was produced by the participants. The question which participants had to ask about related to general information about the agent (e.g. “Do you have sisters or brothers?”) and were taken in part from the small talk questions described in Aron et al. (1997). After 8 min, participants took off the HMD, answered rating questions, and filled in the social presence subscale of the MPS.

Afterwards, participants put on the HMD again and the conversational task “knowledge test” was started in which participant had to answer short questions on world knowledge (e.g. “What is the highest building in the world?”). Before the task, participants were informed about the task, and it was highlighted that they were free to ask the virtual agent for help. In contrast to the previous task, participants had to activate agent’s response mode (where the LLM generated responses for input) by pressing a button. The response mode was automatically deactivated after the agent/LLM had responded. Questions alternated between questions that were relatively easy and relatively hard to answer (based on the experimenter’s evaluation). Participants had to give their answer verbally and could then forward to the next question using the motion controller. The knowledge test lasted for 8 min and was followed by ratings and the social presence subscale of the MPS outside Virtual Reality.

“Convincing” was conducted as a last conversational task where participants had to ask the virtual agent about preferences regarding two options (e.g. “What do you prefer: cats or dogs?”). Following the agent’s response, participants then had to convince the agent of the alternative that had not been chosen by the agent (also regardless of their own preferences). In addition, participants were instructed that they were free in how many attempts they would try to convince the virtual agent. Preference questions were displayed on the virtual screen and participants could forward to the next question using the motion controller. The conversational task had a duration of 8 min and was followed by a final set of ratings and the social presence questionnaire. The experiment had a duration of 40 min, which included about 30 min in VR and 15 min outside VR where the ratings were collected.

After the last conversational task participants were asked to fill in questionnaires regarding their general attitudes towards AI (GAAT), and personality (NEO-FFI 30). Duration of the final part was approximately 20 min. The study had a total duration of about 1.5 h (including onboarding and debriefing).

2.5. Pre-test: validation of LLM persona

To ensure that the personality manipulation of the LLM resulted in the intended changes a personality test (NEO-FFI-30) was conducted for each persona prompt. To account for variability in model output, the questionnaire was administered 100 times with both personas. Reliability was calculated across repetitions using Cronbach’s alpha. Consistency was estimated at .87 for the extraverted persona and at .92 for the introverted persona demonstrating good reliability for both personas. Personas were compared for each NEO-FFI subscale. Results revealed that, across repetitions, the extraverted persona scored significantly higher than the introverted on the extraversion subscale (extraverted: $M = 12.81$, $SD = 4.21$, introverted: $M = 7.87$, $SD = 3.09$), $t(181.78) = 9.64$, $p < .001$, $d = 1.36$, while there were no differences on the neuroticism (extraverted: $M = 9.38$, $SD = 3.50$, introverted: $M = 9.39$, $SD = 2.96$), $t(192.85) = -.02$, $p = .983$, $d < .01$, conscientiousness (extraverted: $M = 9.83$, $SD = 3.51$, introverted: $M = 9.30$, $SD = 3.04$), $t(194.02) = 1.14$, $p = .510$, $d = .16$, and openness subscale (extraverted: $M = 8.46$, $SD = 3.81$, introverted: $M = 7.51$, $SD = 3.26$), $t(193.39) = 1.90$, $p = .178$, $d = .27$. On the agreeableness subscale the introverted persona scored higher than the extraverted persona (extraverted: $M = 5.44$, $SD = 3.09$, introverted: $M = 7.70$, $SD = 3.01$), $t(197.88) = -5.24$, $p < .001$, $d = .74$. Overall, the largest effect of persona was observed with respect to extraversion.

2.6. Data Preprocessing and Statistical analysis

The present study quantified participants’ number of words, number of turns, and number of questions asked in order to quantify interactive engagement, i.e. the degree to which a participant engaged in the conversation. Therefore, participants’ speech was transcribed using the speech-to-text model (see above) and written to a logfile. These data were analyzed by counting the number of words and turns (e.g. an uninterrupted segment of speech of the participant with no interleaved response from the virtual agent) for each question during a conversational task. Note that to characterize interactive engagement, words and turns were analyzed per question rather than as a total number. This was necessary because otherwise simply asking the displayed question and then switching to the next question without any true interactive engagement would have increased the total number of words or turns (and this behavior would have been more likely to occur for the introverted persona where responses were short). One participant was excluded from the analysis because only one question was asked.

Statistical analysis was conducted in the R environment (R Core Team, 2016). Data for each dependent variable were analyzed using a mixed effect ANOVA with persona as between-subject and conversational task as a within-subject factor. Sphericity violations were corrected

using Greenhaus-Geisser correction. In case of significant interaction effects post-hoc t -test were conducted using Holm method to correct for multiple comparisons. Alpha level was set at .05.

2.7. Open Science Statement

Study procedures and hypotheses were preregistered (<https://osf.io/tukqh>). Ratings, questionnaires, and secondary data on conversations (word number, turns) as well as analysis scripts are publicly available in an online repository (<https://osf.io/ws7jf/>). Note that transcriptions of conversations may contain personal information and are, therefore, not publicly available.

3. Results

3.1. Manipulation check

The number of words generated by the LLM per turn was compared between personas as a manipulation check. The extraverted persona generated significantly more words ($M = 24.19$, $SD = 75.62$) per turn than the introverted persona ($M = 15.35$, $SD = 29.80$), $t(41.90) = 11.08$, $p < .001$, $d = 3.30$ and was thus significantly more talkative. These data demonstrate that the personality manipulation via persona prompts was successful and resulted in the intended effects (see also validation of LLM persona above).

3.2. Social evaluation of virtual agent

We investigated how interacting with different LLM personas during different conversational tasks affected participants’ social evaluation of the virtual agents in terms of sympathy and closeness (Fig. 2).

Analysis of sympathy of the virtual agent revealed a main effect of persona, $F(1,44) = 21.57$, $p < .001$, $\eta_p^2 = .33$, a main effect of conversational task, $F(2, 88) = 9.82$, $p < .001$, $\eta_p^2 = .18$, but no interaction effect between persona and conversational task, $F(2, 88) = 2.97$, $p = .056$, $\eta_p^2 = .06$. Sympathy was higher for extraverted ($M = 58.20$, $SD = 12.12$) than for introverted persona ($M = 41.19$, $SD = 12.71$). Sympathy was rated higher for the knowledge test task ($M = 56.70$, $SD = 12.37$) than for small talk ($M = 45.91$, $SD = 21.31$), $t(45) = 3.77$, $p = .001$, and convincing task ($M = 46.47$, $SD = 46.48$), $t(45) = 3.80$, $p = .001$, but

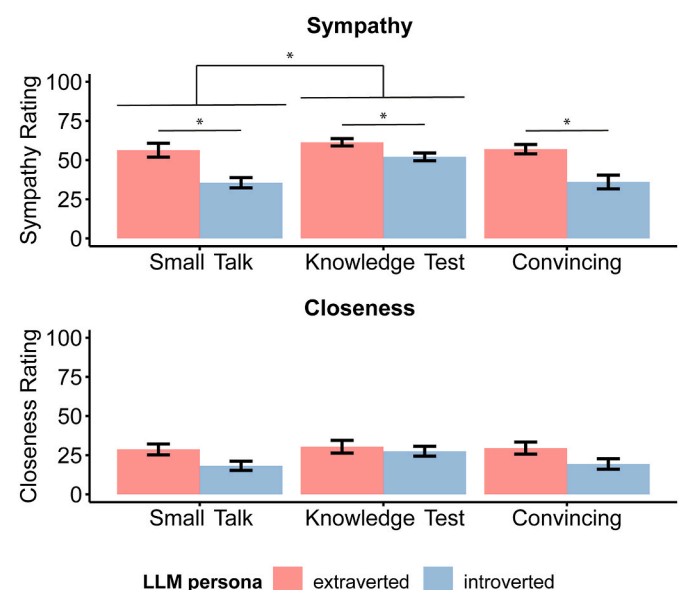


Fig. 2. Social evaluation of the virtual agent in ratings of sympathy (top) and perceived closeness (bottom) as a function of LLM persona and conversational task. Error bars reflect standard error of the mean.

there was no significant difference between small talk and convincing, $t(45) = -.20$, $p = .843$.

Analysis of participants' feelings of closeness towards the virtual agent revealed a main effect of *conversational task* $F(2, 88) = 3.69$, $p = .029$, $\eta_p^2 = .08$, but no main effect of *persona*, $F(1, 44) = 3.36$, $p = .074$, $\eta_p^2 = .07$, and no interaction effect, $F(2, 88) = 1.96$, $p = .146$, $\eta_p^2 = .04$. Follow-up t -tests, however, did not show any significant differences between conversational tasks (all $p > .05$ after correction for multiple comparisons).

Overall, the extraverted persona was rated as more likable than the introverted persona and agents were rated as most likable during the knowledge test. Closeness was not significantly affected by persona or task.

3.3. Emotional experience

To characterize the influence of LLM persona and conversational task on emotional experience valence and arousal ratings were analyzed (Fig. 3).

ANOVA results for valence ratings revealed a main effect of *persona*, $F(1, 44) = 6.08$, $p = .018$, $\eta_p^2 = .12$, but no main effect of *conversational task*, $F(2, 88) = .34$, $p = .714$, $\eta_p^2 < .01$, and no interaction effect between *persona* and *conversational task*, $F(2, 88) = .33$, $p = .720$, $\eta_p^2 < .01$. Participants interacting with the extraverted persona rated their experience as more pleasant ($M = 56.97$, $SD = 13.64$) than participants interacting with the introverted persona ($M = 45.01$, $SD = 18.88$), but experience was not influenced by conversational task.

Analysis of arousal ratings revealed only a main effect of *conversational task*, $F(2, 88) = 5.49$, $p = .006$, $\eta_p^2 = .11$, but no main effect of *persona*, $F(1, 44) = 1.03$, $p = .316$, $\eta_p^2 = .02$, and no interaction effect, $F(2, 88) = 1.52$, $p = .224$, $\eta_p^2 = .03$. Participants rated both the knowledge test task ($M = 44.96$, $SD = 22.58$) and the convincing task ($M = 44.71$, $SD = 20.77$) as more arousing than the small talk task ($M = 35.24$, $SD = 20.07$), $t(45) = 2.76$, $p = .022$ and $t(45) = 2.81$, $p = .022$ respectively. There was no difference in arousal ratings between the knowledge test and the convincing task, $t(45) = .07$, $p = .041$.

LLM persona and conversational tasks had a differential effect on emotional experience, while interacting with a extraverted persona increased pleasantness compared to interacting with an introverted persona, arousal was mainly driven by conversational task, with convincing and knowledge test being more arousing than the small talk task.

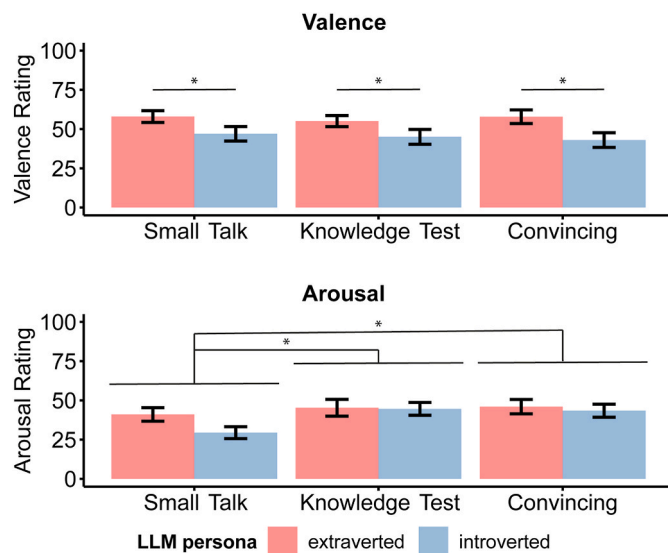


Fig. 3. Participants' emotional experience during social interaction in terms of valence (top) and arousal (bottom) as a function of LLM persona and conversational task. Error bars reflect standard error of the mean.

3.4. Experience of virtual interactions

Another goal was to investigate how LLM persona and conversational task influenced how realistically the interactions with the virtual agent were rated and how social presence was experienced (Fig. 4).

ANOVA of the realism ratings revealed a main effect of *conversational task*, $F(2, 88) = 6.89$, $p = .002$, $\eta_p^2 = .14$, but no main effect of *persona*, $F(1, 44) = 3.43$, $p = .071$, $\eta_p^2 = .07$. However, there was a significant interaction effect of *persona* and *conversational task*, $F(2, 88) = 5.07$, $p = .008$, $\eta_p^2 = .10$. Post-hoc Welch t -tests revealed that interaction with the extraverted persona ($M = 38.26$, $SD = 20.14$) was rated as more realistic than interaction with the introverted persona ($M = 20.52$, $SD = 17.07$) during the convincing task, $t(42.85) = 3.22$, $p = .002$, but there was no difference in realism between personas for the small talk (Extraverted: $M = 26.91$, $SD = 16.38$; Introverted: 21.78 , $SD = 16.13$), $t(43.99) = 1.07$, $p = .290$, and the knowledge test task (Extraverted: $M = 35.09$, $SD = 23.99$; Introverted: 32.61 , $SD = 15.38$), $t(37.46) = .41$, $p = .679$. A follow-up on the main effect of *conversational task* showed that the knowledge test was rated as more realistic than the small talk task, $t(45) = 3.38$, $p = .004$, but there were no differences between the small talk and convincing task, $t(45) = -1.78$, $p = .129$, and between the knowledge test and convincing task, $t(45) = 1.90$, $p = .129$. Overall, persona affected the experienced realism of an interaction only in the convincing task, and interaction during the knowledge test was rated as more realistic than small talk interaction.

In addition, ANOVA results of social presence revealed a main effect of *conversational task*, $F(2, 88) = 6.53$, $p = .002$, $\eta_p^2 = .13$, but no main effect of *persona*, $F(1, 44) = 1.19$, $p = .282$, $\eta_p^2 = .03$, and no interaction effect, $F(2, 88) = .14$, $p = .867$, $\eta_p^2 < .01$. Post-hoc t -test revealed that social presence was significantly higher during small talk compared to the knowledge test, $t(45) = 3.72$, $p = .002$, but there was no significant difference between small talk and convincing, $t(45) = 1.74$, $p = .137$, and between knowledge test and convincing, $t(45) = -1.87$, $p = .137$.

The results show that the extraverted compared to introverted LLM persona increased realism ratings specifically for the convincing task. Interestingly, with respect to conversational tasks, the knowledge test was rated as more realistic than small talk but small talk induced a greater feeling of social presence than the knowledge test.

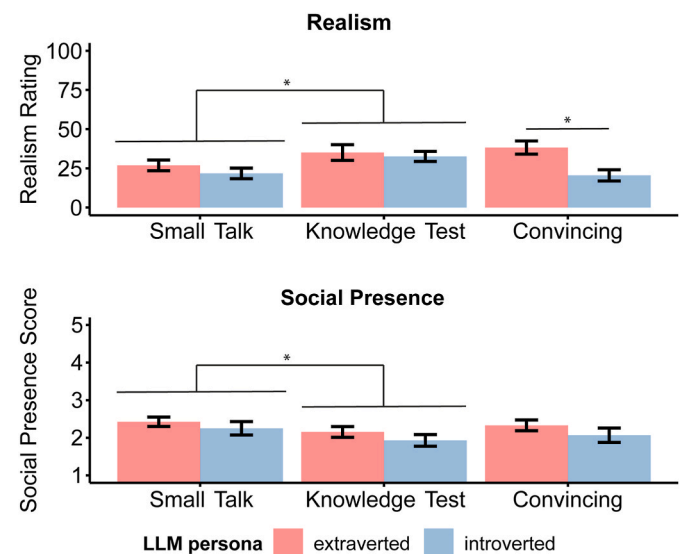


Fig. 4. Evaluation of realism (top) and social presence (bottom) of the virtual interactions as a function of LLM persona and conversational task. Error bars reflect standard error of the mean.

3.5. Behavioral engagement during interaction

In addition to self-reports, behavioral parameters were measured to investigate how LLM persona and conversational task affect participants' engagement in the interaction with the virtual agent (Fig. 5). Three parameters were assessed for this: (1) The number of pre-defined topics/questions that participants worked through during a conversational task. Note, that a higher number means that participants were faster to switch to the next topic and therefore engaged less with the virtual agent for a given topic. (2) The number of words per topic measured the average number of words that participants spoke during a topic, more words means that participants were more engaged in the conversation. (3) The number of initiated turns per topic measured how many back-and-forth alternations with the virtual agent were initialized by participants for a single topic in average. A higher number of turns indicates more interactive engagement.

First, the number of the pre-defined topics/questions that participants worked through with the virtual agent was analyzed. ANOVA results revealed a significant main effect of *persona*, $F(1, 43) = 26.99$, $p < .001$, $\eta_p^2 = .39$, a significant main effect of *conversational task*, $F(1.52, 65.36) = 178.08$, $p < .001$, $\eta_p^2 = .81$, but not interaction effect, $F(1.52, 65.36) = .54$, $p = .538$, $\eta_p^2 = .01$. Participants reached more of the pre-defined topics/questions when interacting with the introverted ($M = 14.11$, $SD = 7.01$) compared to the extraverted persona ($M = 10.06$, $SD = 6.34$). The highest number of topics/questions was reached in the knowledge test task ($M = 19.11$, $SD = 5.45$), followed by the social interaction task ($M = 11.24$, $SD = 4.13$), and the convincing task ($M = 5.76$, $SD = 2.89$). Post-hoc t-tests showed significant differences between all conversational tasks ($ps < .001$). Overall, participants reached a higher number of topics/questions when interactions were shorter and when the persona manipulation resulted in shorter responses of the virtual agent (i.e. in the introverted condition).

Next, the average number of words per pre-defined topic/question was analyzed. The ANOVA revealed an effect of *persona*, $F(1, 43) = 16.43$, $p < .001$, $\eta_p^2 = .28$, and *conversational task*, $F(1.26, 54.29) =$

168.17 , $p < .001$, $\eta_p^2 = .80$, as well as an interaction between *persona* and *conversational task*, $F(1.26, 54.29) = 6.96$, $p = .007$, $\eta_p^2 = .14$. Post-hoc t-tests were conducted to follow-up on the interaction effect. There was a significant effect of *persona* for each level of conversational task, with greater number of words for the extraverted persona compared to the introverted persona (Small Talk: $t(30.84) = 2.92$, $p = .013$; Knowledge Test: $t(30.53) = 2.43$, $p = .021$; Convincing: $t(42.01) = 3.34$, $p = .005$). However, an analysis of differences between conversational tasks revealed that the extraverted persona group produced significantly more words in the convincing than the knowledge task compared to the introverted persona group, $t(42.11) = 3.12$, $p = .010$, but groups did not differ in differences between the small talk and knowledge test task, $t(31.67) = 2.14$, $p = .067$, and differences between the small talk and convincing task, $t(39.76) = -2.20$, $p = .068$. Overall, participants spoke more words with the extraverted persona agent than the introverted persona agent, and more words in the convincing task than the small talk and knowledge test task. Finally, participants talking with the extraverted persona specifically increased the number of words in the convincing task relative to the knowledge task compared to participants talking to the introverted persona agent.

In a further analysis, the number of turns initiated by the participant per topic/question was investigated. ANOVA results showed a significant main effect of *persona*, $F(1, 43) = 4.36$, $p < .001$, $\eta_p^2 = .09$, a significant main effect of *conversational task*, $F(1.39, 59.57) = 111.62$, $p < .001$, $\eta_p^2 = .72$, but there was no significant interaction effect, $F(1.39, 59.57) = 1.16$, $p = .304$, $\eta_p^2 = .03$. Participants interacting with the extraverted persona initiated more turns per question ($M = 2.78$, $SD = 2.17$) than participants interacting with the introverted persona ($M = 2.27$, $SD = 1.70$). With respect to conversational task, participants initiated more turns per question in the convincing task ($M = 4.40$, $SD = 1.95$) than in the small talk task ($M = 2.46$, $SD = 1.00$), $t(44) = 6.66$, $p < .001$, and more turns per question in the small talk compared to knowledge test task ($M = .72$, $SD = .24$), $t(44) = 11.85$, $p < .001$. Therefore, participants were more likely continue a conversation about a particular topic when interacting with the extraverted compared to the introverted agent and there the most attempts to continue the conversation were observed in the convincing task followed by the small talk task.

Overall, the behavioral parameters show a consistent pattern, with the extraverted LLM persona resulting in more interactive engagement than the introverted persona. This effect was more pronounced in the convincing task for the number of words but not the number of turns.

3.6. Using AI for help in the knowledge test

During the knowledge test, participants had to answer questions and were free to ask the virtual agent for help. Questions were grouped into easy and difficult questions. The percentage of questions for which participants asked for help was then analyzed with a mixed ANOVA using the between-subject factor *persona* and the within-subject factor *question difficulty*. The analysis revealed a significant main effect of *difficulty*, $F(1, 44) = 666.49$, $p < .001$, $\eta_p^2 = .94$, but no effect of *persona*, $F(1, 44) = 1.01$, $p = .321$, $\eta_p^2 = .02$, and no interaction effect between *persona* and *difficulty*, $F(1, 44) = .28$, $p = .601$, $\eta_p^2 < .01$. Therefore participants asked the LLM-controlled agent more frequently for help when difficult questions had to be answered ($M = 90.03\%$, $SD = 9.91\%$) compared to when easy questions had to be answered ($M = 23.12\%$, $SD = 17.47\%$), but asking for help was not influenced by persona of the LLM controlled virtual agent (see Fig. 6).

In addition, participants rated how confident they were about the correctness of their answers, either for questions answered with or without help of the virtual agent (factor *help*). Analysis of these confidence ratings revealed a main effect of *help*, $F(1, 44) = 11.19$, $p = .002$, $\eta_p^2 = .20$, but no effect of *persona*, $F(1, 44) = .71$, $p = .404$, $\eta_p^2 = .02$, and no interaction effect, $F(1, 44) = .11$, $p = .740$, $\eta_p^2 < .01$. Participants were more confident about the correctness of their answers when they

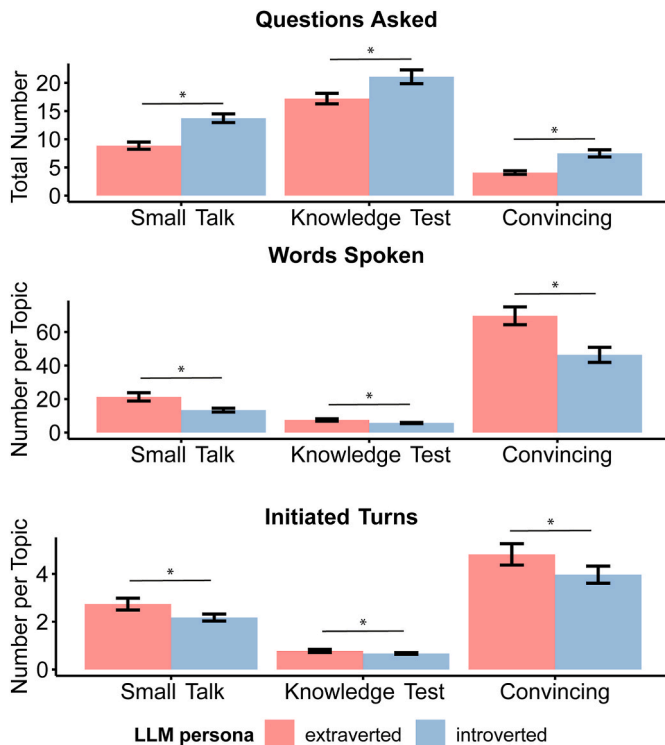


Fig. 5. Behavioral parameters during conversations as a function of LLM persona and conversational task. Error bars reflect standard error of the mean.

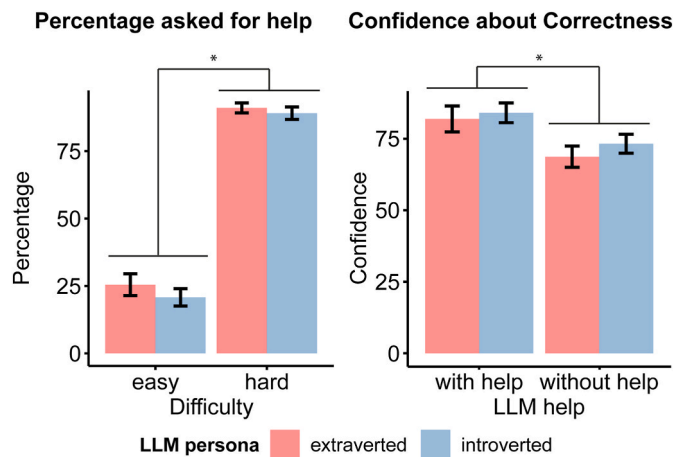


Fig. 6. Behavior and Confidence in the knowledge test conversational task. Graph left shows the percentage of answered questions for which participants asked the virtual agent for help as a function of question difficulty and persona. Graph right shows how participants rated their confidence in the correctness of their answer as a function of whether they had asked the virtual agent for help and persona. All error bars reflect the standard error of the mean.

asked the LLM controlled agent for help ($M = 82.89$, $SD = 19.07$) compared to when they answered the question without help ($M = 70.12$, $SD = 16.80$) but this was not modulated by personality of the virtual agent.

Overall, the data show that participants were more likely to ask the LLM for help when hard compared to easy questions had to be answered and they were more confident in the answers when they had asked for help, however, there was no influence of the LLM persona.

4. Discussion

The goal of this study was to investigate how the persona of an LLM controlled embodied conversational agent modulates experience and behavior in social interactions across different conversational tasks. Participants interacted with either an extraverted or an introverted LLM persona during a small talk, a knowledge test or a convincing task. In order to characterize the influence of these manipulations we assessed social-evaluative processes, emotional experience, and evaluation of the VR scenario, as well as behavioral parameters such as interactive engagement and participants' tendency to ask the LLM for help. This design allowed to test whether participants would form social evaluations of the LLM agents similar to real-life encounters (CASA) and whether this would also result in an alignment between verbal communicative style between persons and the LLM agents. Overall, we found consistent evidence for an effect of persona in line with our hypotheses that conversing with the extraverted persona resulted in a more positive social evaluation, a more positive emotional experience and greater interactive engagement than conversing with the introverted persona. There was, however, no consistent interaction effect between LLM persona and conversational tasks. The effect of LLM persona was only modulated by conversational tasks with respect to the experienced realism and number of words, in a sense that the relative increase in realism and engagement from introverted to extraverted was most pronounced in the convincing task. This finding did not confirm the hypothesis that the effect of extraversion would be greater in the small talk compared to the knowledge test and convincing task. Furthermore, in contrast to our hypothesis the small talk was not more pleasant than the knowledge test and convincing task. Finally, while we found no support for the hypothesis that a extraverted LLM personality would increase participants tendency to ask the virtual agent for help during the knowledge test, we did find that raising question difficulty strongly increased the tendency to ask for help and that participants were more

confident in the answers when they had asked the LLM controlled agent for help. Overall, the present study demonstrates that persona of a LLM embodied conversational agent strongly influences experience and behavior in social interactions, although more data are required to characterize the influence of conversational tasks.

Prompting persona traits in a LLM strongly affects social evaluation of the virtual agent, emotional experience, and interactive engagement. Interaction with the extraverted persona increased the feeling of sympathy towards the virtual agent compared to the introverted persona, similarly as in real-word interactions (Wortman & Wood, 2011). In line with findings from chat-based interactions (Liu & Sundar, 2018) our results demonstrate that users are sensitive to the communicative style of a virtual interactive agent and adapt their behavior and socio-emotional processing accordingly. Our findings therefore extend the literature on LLM personality (Frisch & Giulianelli, 2024; Jiang et al., 2024; Serapio-García et al., 2023) with direct evidence on effects of LLM personality on user experience. The more positive evaluation of the extraverted persona in comparison to the introverted persona is in line with previous studies which found more positive social evaluations (i.e. liking) in real-life for persons scoring high on sociable aspects of extraversion such as being talkative and joyful (Wortman & Wood, 2011). However, it should be noted that extraversion may have a strong influence on an initial positive evaluation but may have a weaker influence on the formation of social bonds across longer interactions (Harris & Vazire, 2016). Interestingly, this was also reflected in the present study, where we did not observe an influence of LLM persona on interpersonal closeness. However, such deeper social connections may develop over longer periods of time as has been demonstrated for multi-session interactions with a chatbot (Araujo & Bol, 2024). Future studies should therefore implement long-term interactions with LLM-controlled embodied conversational agents to investigate social affiliation and adaptation processes. Overall, the present study demonstrates that manipulations of extraversion via LLM prompts impact socio-evaluative processes of face-to-face social interactions in immersive virtual environments and – in line with the CASA framework – simulate findings from human-human interaction.

A further target of our study was to shed light on the influence of task characteristics on social interaction. Previous studies have highlighted the role of small talk to foster trust and social affiliation in conversations (Epley & Schroeder, 2014; Saffari & Azaria, 2024; Álvaro Aranda & Lázaro Gutiérrez, 2022). In contrast, in the present study participants experienced the argumentation focused conversation as equally pleasant yet more arousing than the small talk conversation. It should be noted, however, that the argumentation focused conversation included topics which are also suitable for small talk conversations (e.g. "What do you prefer: Dogs or cats?") and do not reflect strong controversies. This suggests that the exchange of arguments was experienced as a deeper conversation rather than an adverse disputation and can be explained by the fact that the LLM was trained to be a helpful assistant and therefore responses of the model were always respectful and constructive. Furthermore, we observed a modulation of the impact of persona by task for the evaluation of interactive realism. Interaction with the extraverted persona was experienced as more realistic than interaction with the introverted persona in the deeper argumentative conversations, but persona did not affect realism in the other conversational tasks. This suggests that persona manipulations become more important when more elaborative conversations with agents are implemented. In contrast, the effects of persona on social evaluation as well as emotional experience were consistent across conversational tasks suggesting that participants did not change their initial evaluations. Finally, it should be noted that virtual agents were evaluated as most positive during the knowledge test suggesting that receiving help by a conversational agent per se results in a more favorable evaluation of the agent. Overall, these data demonstrate that task demands and role of an agent can affect the experience of interactions with LLM-controlled agents and can also modulate the effects of agent characteristics.

Notably, the experimental manipulations of persona and task affected not only experience of social interactions, but also had a direct impact on participants' behavior. Interacting with an extraverted compared to an introverted persona agent resulted in greater interactive engagement. While the underlying mechanisms remain to be explored, it is possible that participants' tendency to mimic the agents' communicative style may drive the effect (Frisch & Giulianelli, 2024; Kroczeek & Gunter, 2020; Schoot et al., 2016). This is in line with the interactive alignment model (Pickering & Garrod, 2004) and demonstrates that conversations with LLM-controlled agents result in similar behavioral adaptation in communicative patterns as in human-human conversations – as predicted by the CASA framework (Nass & Moon, 2000). The finding that interacting with a talkative agent modified participants' behavior by making them more talkative themselves also aligns with existing literature on behavioral effects of conversations with LLM agents with respect to health decisions and learning (Lim et al., 2024; Sonlu et al., 2024). Importantly, it should be discussed whether the observed effect is truly a mimicry/alignment effect or whether other explanations can account for the differences in interactive engagement. Note that the quantity of words and turns was neither task-relevant nor instructed. Nevertheless, participants changed their behavior according to the communicative style of the agent, suggesting true alignment. Alternatively, however, it remains possible that the effect was driven by the change in social evaluation and emotional experience and therefore only indirectly related to the communicative style. In this line of argumentation it is possible that participants talked more when they felt more pleasant and when they liked the agent better (Forgas, 2011). In the present study we cannot rule out this alternative, but future studies might control for emotional experience and social evaluation to test mimicry effects. Besides interactive engagement across conversational tasks, the study also investigated participants' tendency to ask the conversational agent for help to answer knowledge questions. Using LLMs to acquire knowledge is a typical use task for human-AI interaction. Our findings did not reveal an effect of persona, neither for easy nor difficult questions. Interestingly, however, participants had greater confidence in their answer when they relied on the conversational agent (cf. Wester et al., 2024). This demonstrates that there is a general tendency to accept the answers provided by a LLM as true (Zhou et al., 2019). Overall, the present study provided evidence that interacting with an virtually embodied and LLM-controlled conversational agent directly impacts users' behavior and this effect can be further modulated by manipulating the agent's characteristics and communicative style.

Social interactions with LLM-controlled embodied virtual agents are a novel and promising way to study social behavior and human-computer interactions. However, there are some limitations that need to be discussed. First, the investigated sample was restricted in age group and cultural backgrounds. As these factors may affect the evaluation of AI agents (Draxler et al., 2023), future studies should be conducted to assess effects of individual differences. Another limitation is the order of tasks. Conversational tasks were presented in a fixed order which may have resulted in carry-over effects as an impression of the agent acquired in the small talk task might have persisted across the other tasks (Bar et al., 2006; Digirolamo & Hintzman, 1997). With this limitation in mind, it should be noted that the order of the present study was set to resemble a natural interaction between humans, where a short phase of getting to know each other typically precedes deeper interaction. Furthermore, we found that some parameters differed only in later conversational tasks, suggesting that task-specific effects were preserved. A further limitation is that the virtual agents did not produce non-verbal behavior or emotional expressions together with the speech output. Non-verbal behaviors have been demonstrated to have a crucial role for communication and social interactive behavior (Kroczeek et al., 2024; Kroczeek & Mühlberger, 2023), future studies should therefore aim at incorporating both non-verbal and verbal behavior like gestures and facial expressions resulting in a more naturalistic social interactions (for an example see Llanes-Jurado et al., 2024). Finally, it should be noted

that while our prompt manipulation resulted in changes in particular personality dimensions, future studies might implement more broadly defined personality manipulations, which would also allow to manipulate similarity between humans and agents across different personality dimensions (Lim et al., 2024).

Our work has important implications for the implementation of LLM-controlled embodied conversational agents in social interactions and shows that simple personality prompts can influence experience and behavior in such interactions. Furthermore, these effects can be observed over a range of conversational tasks, even when personality is task irrelevant. This is relevant for many applications, e.g. in education or health care, which aim at activating users or providing personalized interactions (Shumanov & Johnson, 2021). The impact of LLM-controlled embodied conversational agents on experience and behavior is also promising for VR applications that provide mental health interventions (Bell et al., 2024; Graf et al., 2024; Herbener et al., 2024). Finally, our data support the CASA framework, which has implications on the use of LLM-agents in social settings. People might build strong social relations to LLM agents and treat them as equal interaction partners. This might have positive outcomes with respect to health benefits of social connections (Maples et al., 2024), however, because LLM agents are ultimately controlled by a pre-trained model, agents might be also used for manipulation (Leib et al., 2021) or lead to a devaluation of real world social relationships. Future studies should evaluate these chances and risks.

5. Conclusion

The present study highlights the use of LLMs to control conversations with embodied virtual agents and demonstrates that an agent's persona characteristics and communicative style can have a profound impact on experience and behavior across different conversational tasks that mimics real-life. The data support the hypothesis that conversations with extraverted compared to introverted agents result in more positive social evaluation, emotional experience, and interactive engagement. In contrast to our hypotheses, however, these effects were not driven by one particular conversational task but rather observed across tasks and may thus represent general socio-affective processing. This has implications for the use of embodied conversational agents in applications, for example in mental health, and for research focusing on social behavior.

CRedit authorship contribution statement

Leon O.H. Kroczeek: Supervision, Formal analysis, Writing – original draft, Validation, Methodology, Conceptualization, Visualization, Project administration, Data curation. **Alexander May:** Software. **Selina Hettenkofer:** Investigation. **Andreas Ruider:** Software. **Bernd Ludwig:** Writing – review & editing. **Andreas Mühlberger:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2025.108759>.

Data availability

Data and analysis scripts are available in an data repository <https://osf.io/ws7jf/>.

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